

EXPERIMENT - II

16 BIT MULTIPLICATION.

AIM:

To write an assembly language program to implement 16 bit multiplication on 8086 processor.

ALGORITHM:

1) load the first data in HL pair.

2) move content of HL pair to stack pointer.

3) load the second data in HL pair and move to DE

4) make H as 0H and L as 0H

5) Add HL pair and stack pointer.

6) check for carry if carry increment by 1 else move to next step.

7) move E to A & perform with accumulation and register D.

8) The value of operation is zero the solve the value else go to 3.

PROGRAM:

MNEMONICS

EXPLANATION

MOV AX, 1100H

move A to 1100



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MOV BX, 1102H
MUL BX
MOV [1200H], AX
MOV 1202, DX
HLT

```

move the B to 1102
 multiply with [B]
 move 1200 to A
 move the 1202 to D
 halt.

INPUT:

ADDRESS	DATA
1100	20
1202	3

OUTPUT:

ADDRESS	DATA
1200	60
1202	3

RESULT :

Thus the program was
 executed successfully using 8086
 emulator.

100 [1202H], DX
HLT

Random Access Memory			
0100:1100		update	table list
0100:1100:	20	032	SPA
0100:1101:	00	000	NULL
0100:1102:	00	000	NULL
0100:1103:	00	000	NULL
0100:1104:	00	000	NULL
0100:1105:	00	000	NULL
0100:1106:	00	000	NULL
0100:1107:	00	000	NULL
0100:1108:	00	000	NULL

emulator: noname.bin_

filemathdebugviewexternalvirtual devicesvirtual drivehelp

Load

reload

step back

single step

run

step delay ms: 0

registers

	H	L
AX	00	00
BX	00	00
CX	00	00
DX	00	00
CS	0100	
IP	0010	
SS	0100	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0100	
ES	0100	

0100:0010

01000: A1 161 i

01001: 00 000 NULL

01002: 11 017 ◀

01003: 8B 139 i

01004: 1E 030 ▲

01005: 02 002 0

01006: 11 017 ◀

01007: F7 247 ≈

01008: E3 227 π

01009: A3 163 ú

0100A: 00 000 NULL

0100B: 12 018 ‡

0100C: 89 137 ë

0100D: 16 022

0100E: 02 002 0

0100F: 12 018 ‡

01010: F4 244 ↑

01011: 90 144 é

01012: 90 144 é

01013: 90 144 é

01014: 90 144 é

01015: 90 144 é

0100:0010

MOV AX, [01100h]

MOV BX, [01102h]

MUL BX

MOV [01200h], AX

MOV [01202h], DX

HLT

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original source c...	
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02	MOV BX, [1102H]
03	MUL BX
04	MOV [1200H], AX
05	MOV [1202H], DX
06	HLT
07	
08	
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